

SAI SINDHURA KOLLEPARA

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SUMMARY

Detail-oriented data professional with 3+ years of experience across data analytics, data engineering, and data science in enterprise and research environments. Strong expertise in SQL analytics, ETL development, machine learning, and dashboarding, with hands-on experience processing large-scale datasets and delivering stakeholder-ready insights. Proficient in Python, SQL, cloud platforms, and BI tools, with a solid foundation in statistical analysis, predictive modeling, and automation.

SKILLS

Languages: Python (pandas, NumPy, matplotlib, seaborn, Plotly), SQL (MySQL, SQL Server), R

Data Engineering & ETL: ETL/ELT pipelines, data cleaning and validation, data transformation, data modeling (star and snowflake schemas), Snowflake, AWS Glue, workflow automation

Data Visualization & Reporting: Power BI (DAX, data modeling), Tableau, Plotly Dash, Looker Studio, Excel

Analytics & Statistics: Exploratory Data Analysis (EDA), hypothesis testing, regression analysis, time series analysis and forecasting, predictive analytics, anomaly detection, cohort and retention analysis

Machine Learning: Supervised and unsupervised learning, feature engineering, model evaluation and validation, Scikit-learn, time series forecasting

Natural Language Processing (NLP): Text classification, topic modeling (LDA), sentiment analysis, semantic similarity

Deep Learning & LLMs: PyTorch, TensorFlow, Hugging Face Transformers, BERT, GPT, LLM-based pipelines, generative AI

Cloud & Data Platforms: AWS (Athena, QuickSight, SageMaker, Lambda), Google BigQuery, Jetstream2 HPC

WORK EXPERIENCE

Data Scientist, Kelley's Data Science and AI Lab – Indiana University Bloomington

Jan 2025 – Present

- Analyzed behavioral, sensor, and survey data from 800+ participants using Python and SQL to identify early indicators of depression and anxiety.
- Processed and modeled 10M+ multi-modal records (motion, location, Bluetooth, battery, health metrics) using reproducible pipelines on Jetstream2 HPC, improving data quality and reducing processing time by 40%.
- Developed statistical and ML-based approaches including correlation analysis, anomaly detection, forecasting, and behavioral trend modeling.
- Built 5+ interactive dashboards (Plotly Dash, Tableau) visualizing retention trends, symptom progression, user-level behavior, and demographic comparisons.
- Delivered insights on engagement patterns, wellbeing trends, and data quality gaps to research leadership and clinicians, supporting study monitoring and decision-making.

Product Intern — AI-Enhanced Underwriting, NeuralMetrics, Colorado

Sep 2025 – Dec 2025

- Improved the GIA Lite AI underwriting assistant by optimizing LLM context and reducing token usage, preserving output accuracy.
- Designed and implemented summary and context-pruning routines to streamline LLM inputs and enhance model responsiveness.
- Supported evaluation and improvements of LLM-powered risk classification workflows used in commercial insurance underwriting.

Data Analyst, Wipro Limited, India

Dec 2021 – Aug 2023

- Assessed 2M+ digital transactions using SQL and AWS Athena to identify system bottlenecks, recurring issues, and usage anomalies, improving reliability of reporting.
- Designed 10+ Power BI dashboards to monitor key performance indicators (KPIs), reducing ad hoc reporting requests by 35%.
- Automated rule-based anomaly detection scripts in Python/SQL to automatically flag abnormal spikes in failures or latency, reducing unresolved issues by 30% and improving resolution times by 50%.
- Collaborated with cross-functional teams to define performance metrics, ensuring consistency in reporting across stakeholders.
- Streamlined 4+ recurring reporting workflows in Python, cutting manual reporting time by 40% and improving output accuracy.

ACADEMIC PROJECTS

Reddit Comments to Post Relevance Analysis | Python, LLMs, Topic Modeling, Hugging Face

- Built a classification pipeline using LDA, BERT, and MiniLM to evaluate comment relevance on 500K+ Reddit threads, improving accuracy by 20%.
- Devised a semantic scoring system and applied K-Means clustering to organize Reddit threads, boosting label quality and reducing moderation by 50%.

Global Emissions Explorer – Time-Series Dashboard | Python, Plotly, Dash

- Built an interactive web dashboard visualizing global greenhouse gas emissions, policy impacts, and historical temperature trends using data from 5+ public sources.
- Implemented interactive filters, animated maps, and KPI tracking to support environmental research and policy analysis.

Global Greenhouse Gas Analysis: Total & Per-Capita Emissions | Python, SQL, BigQuery, Plotly, Looker Studio

- Orchestrated ETL workflows consolidating emissions data from 5+ sources and modeled per-capita emissions across 200+ countries.
- Delivered animated choropleth maps and reporting dashboards via Plotly and Looker Studio, increasing stakeholder engagement by 30%.

Stack Overflow Developer Trends Dashboard | Power BI, DAX, SQL

- Analyzed 100K+ survey responses to uncover trends in developer roles, tools, employment types, and technology preferences.
- Optimized DAX calculations and filters, reducing dashboard load time by 40% and improving drill-down performance.

Enrollment Insights Dashboard | Power BI, DAX, SQL, Excel

- Analyzed institutional enrollment, capacity, and retention trends using university administrative data.
- Engineered DAX metrics for utilization, retention, and year-over-year growth to support data-driven planning.

EDUCATION

Master of Science in Data Science, Indiana University Bloomington **Aug 2023 - May 2025**
Coursework: Data Visualization, Advanced Database Concepts, Applied ML, Data Mining, Cloud Computing-AWS, Statistics

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate (PL-300), AWS Certified Cloud Practitioner